# Requirements: data-driven camera calibration for light field camera systems

**Author:** Youssef Daoudi El Boukhrissi (4277022)

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# Requirements for the Cross-Shaped Camera Array Calibration

Based on the implemented calibration framework and the applied LiFCal-based methodology, the following requirements define the functional scope and measurable performance goals of the system developed in this Bachelor's thesis. The system assumes a predominantly static scene during calibration and monitoring while allowing different camera array poses across captured image sets.

## Functional Requirements

- **FR1 – Initial Reference Calibration (Target-based):**
  The system shall perform a one-time, checkerboard-based initial calibration using OpenCV to estimate intrinsic and extrinsic parameters for the cross-shaped camera array. This initial calibration shall be stored as the baseline calibration and serve as the reference state for all subsequent health evaluations and recalibration decisions.

- **FR2 – Parameter Export and Standardized Representation:**
  The system shall export all calibration results (initial calibration and LiFCal-based recalibration) in a standardized, machine-readable JSON format, including intrinsic parameters, extrinsic poses, reprojection error statistics, timestamps, and metadata.

- **FR3 – Periodic Calibration Health Monitoring:**
  The system shall periodically evaluate the calibration quality by computing a global health score based on the currently active calibration parameters. The monitoring process shall operate automatically and independently of user interaction.

- **FR4 – Health Score Computation (Global Scalar Metric):**
  The system shall compute a single scalar health score in the range $[0, 100]$ representing the overall calibration quality of the complete camera array. For the initial calibration, the health score shall be derived from reprojection error statistics. For LiFCal-based recalibration, the health score shall be derived from LiFCal-specific reprojection and consistency metrics extracted from the combined parameter set.

- **FR5 – Target-Free Online Recalibration (LiFCal-based):**
  The system shall support target-free recalibration using LiFCal based on focus images and virtual depth data generated from recent image sets. The recalibration process shall not require calibration targets and shall operate on automatically prepared image sets.

- **FR6 – Acceptance and Baseline Update Rule:**
  The system shall compare the health score of a newly computed calibration against the health score of the currently stored baseline calibration. The new calibration shall be accepted and stored as the new baseline only if its health score is strictly higher than the baseline score; otherwise, the system shall retain the previous baseline calibration.

- **FR7 – Automated Periodic Recalibration Scheduling:**
  The system shall support automated, periodic execution of the LiFCal-based recalibration process at a configurable time interval (e.g. every 5 minutes). Each

recalibration cycle shall be executed independently and evaluated using the defined health score mechanism.

- **FR8 – Logging and Traceability:**
  The system shall log all calibration and recalibration runs, including timestamps, used image sets, computed health scores, baseline update decisions, and error states. This information shall be stored persistently to support evaluation, debugging, and reproducibility.

## Non-Functional Requirements

- **NFR1 – Modularity:**
  The system shall be structured into modular components (initial calibration, LiFCal recalibration, health monitoring, scheduling, and logging) to support extensibility and maintainability.

- **NFR2 – Reproducibility:**
  All calibration results and health evaluations shall be reproducible using stored JSON files and logged metadata.

- **NFR3 – Robustness:**
  The calibration and monitoring pipeline shall tolerate moderate variations in scene content and illumination without system failure, even if recalibration quality temporarily degrades.

- **NFR4 – Automation:**
  The calibration framework shall operate fully automatically after initial setup, without requiring manual intervention during periodic monitoring and recalibration.